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(54) **DYNAMIC FOCUS FOR LASER PROCESSING HEAD**

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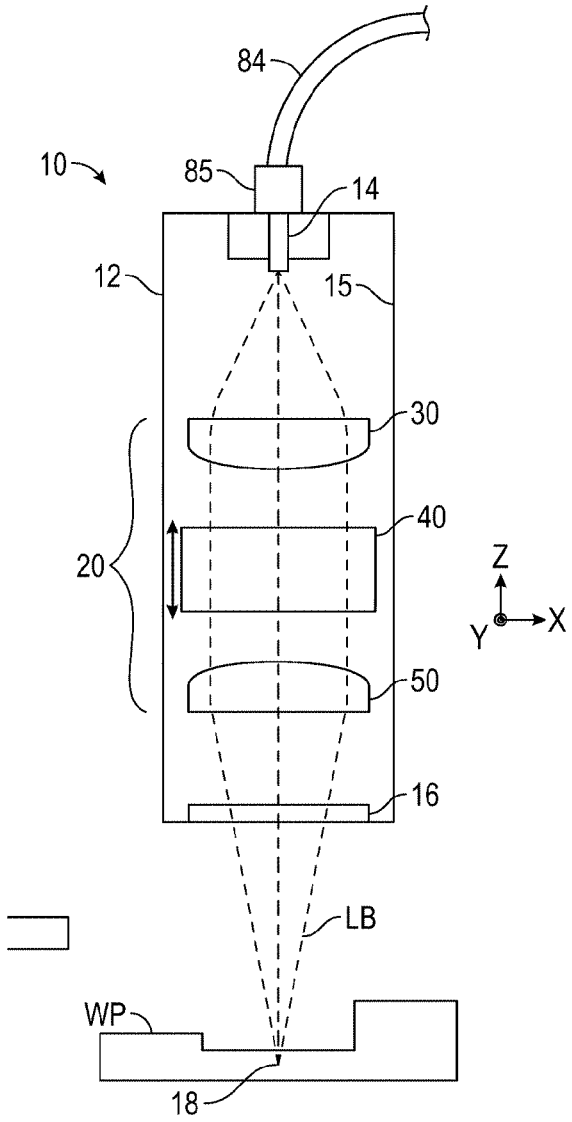
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(52) **U.S. Cl.**

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(57) **ABSTRACT**

A laser processing head transmits a laser beam to a process zone. An input lens fixedly mounted in the head images the beam from an input with a first (negative) focal length along an optical axis. An intermediate lens is movably mounted in the head in connection with an actuator, which can move the intermediate lens along the axis at a variable lens distance relative to the input lens. The intermediate lens images the beam with a second (positive) focal length. An output lens fixedly mounted in the head images the beam with a third (positive) focal length along the axis to a focal point. The third focal length is greater than the second focal length and provides a large working distance. The focal point can be varied at a variable focal distance along the axis in relation to the variable lens distance by which the intermediate lens is moved.



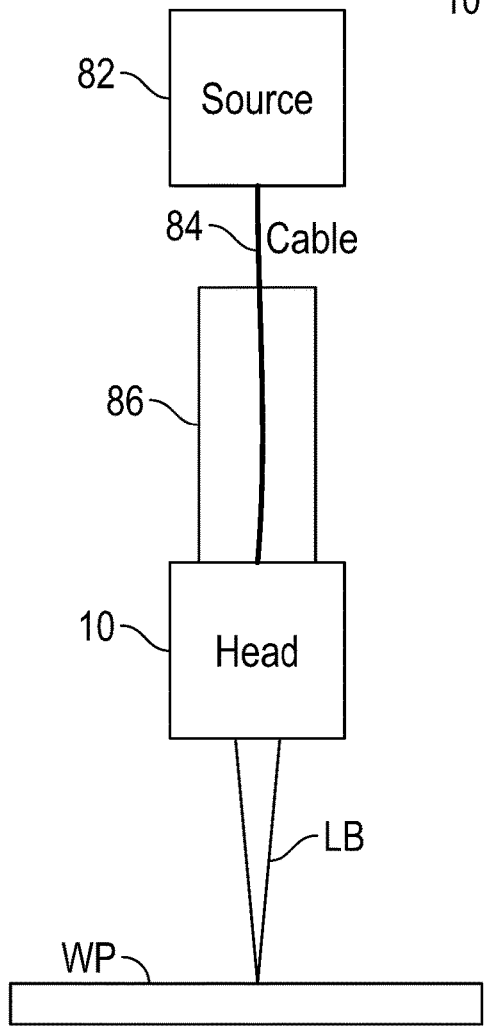


FIG. 1A

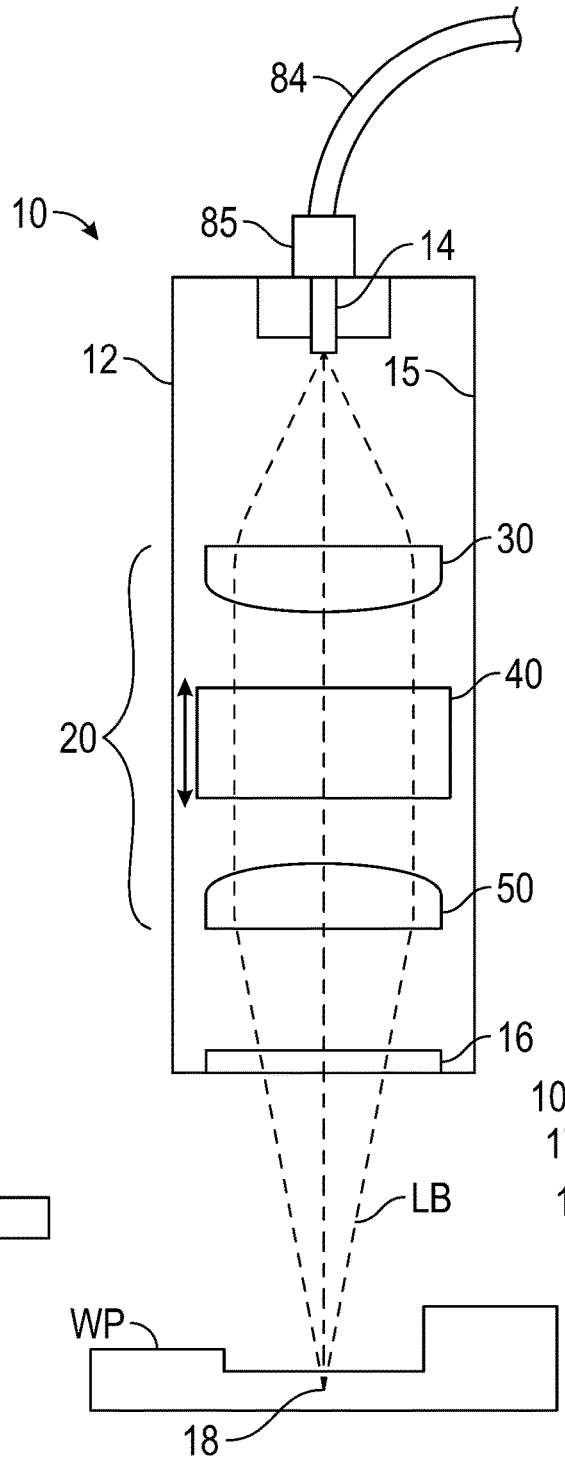


FIG. 1B

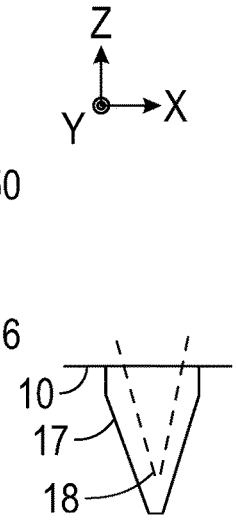


FIG. 1C

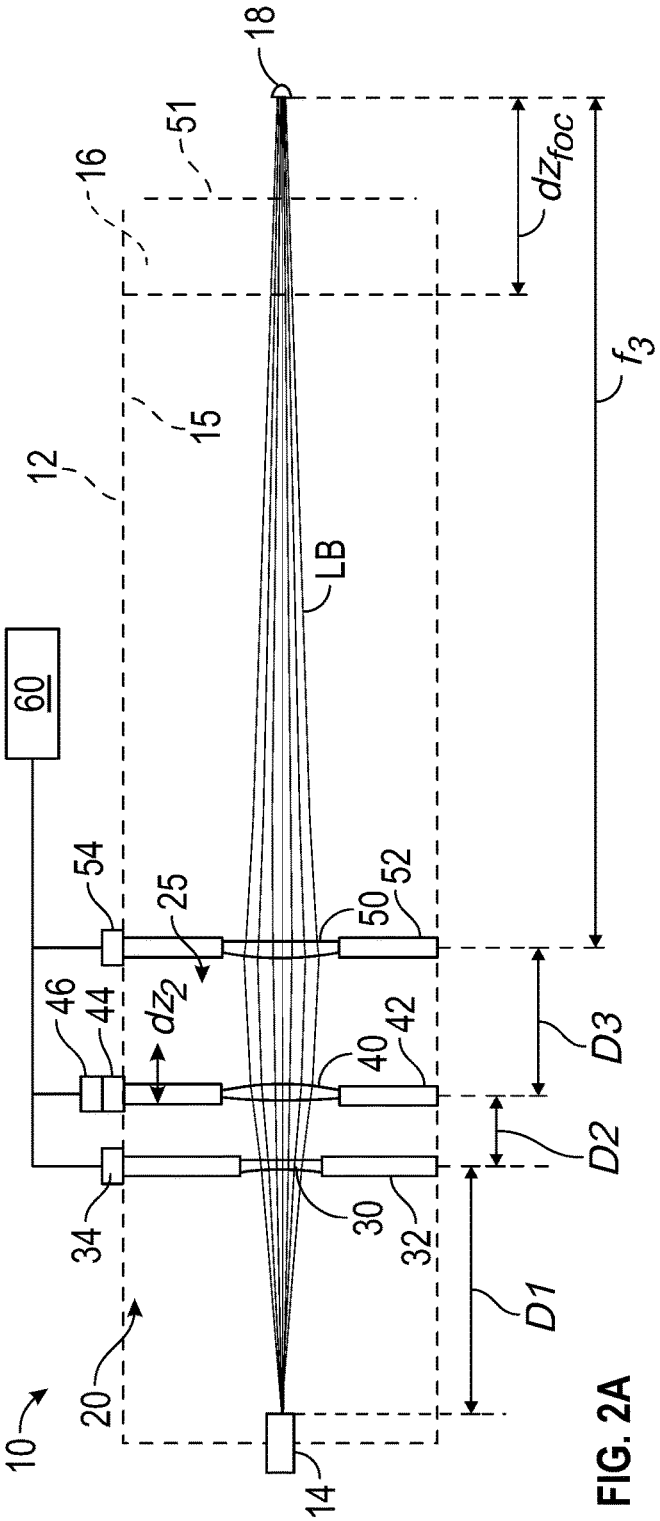

18
51
16
dz_{foc}
f_3
15
12
LB
60
54
25
50
52
46
44
dz_2
40
42
D3
D2
34
30
32
D1
20
10
14

FIG. 2A

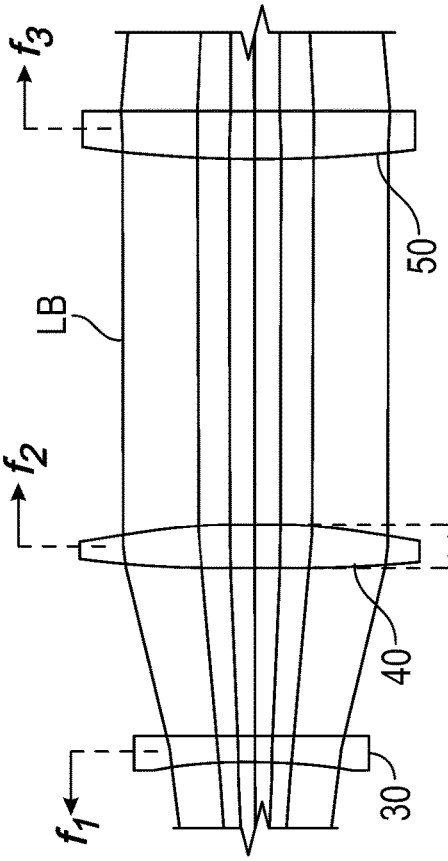

f_3
50
LB
f_2
dz_2
40
f_1
30

FIG. 2B

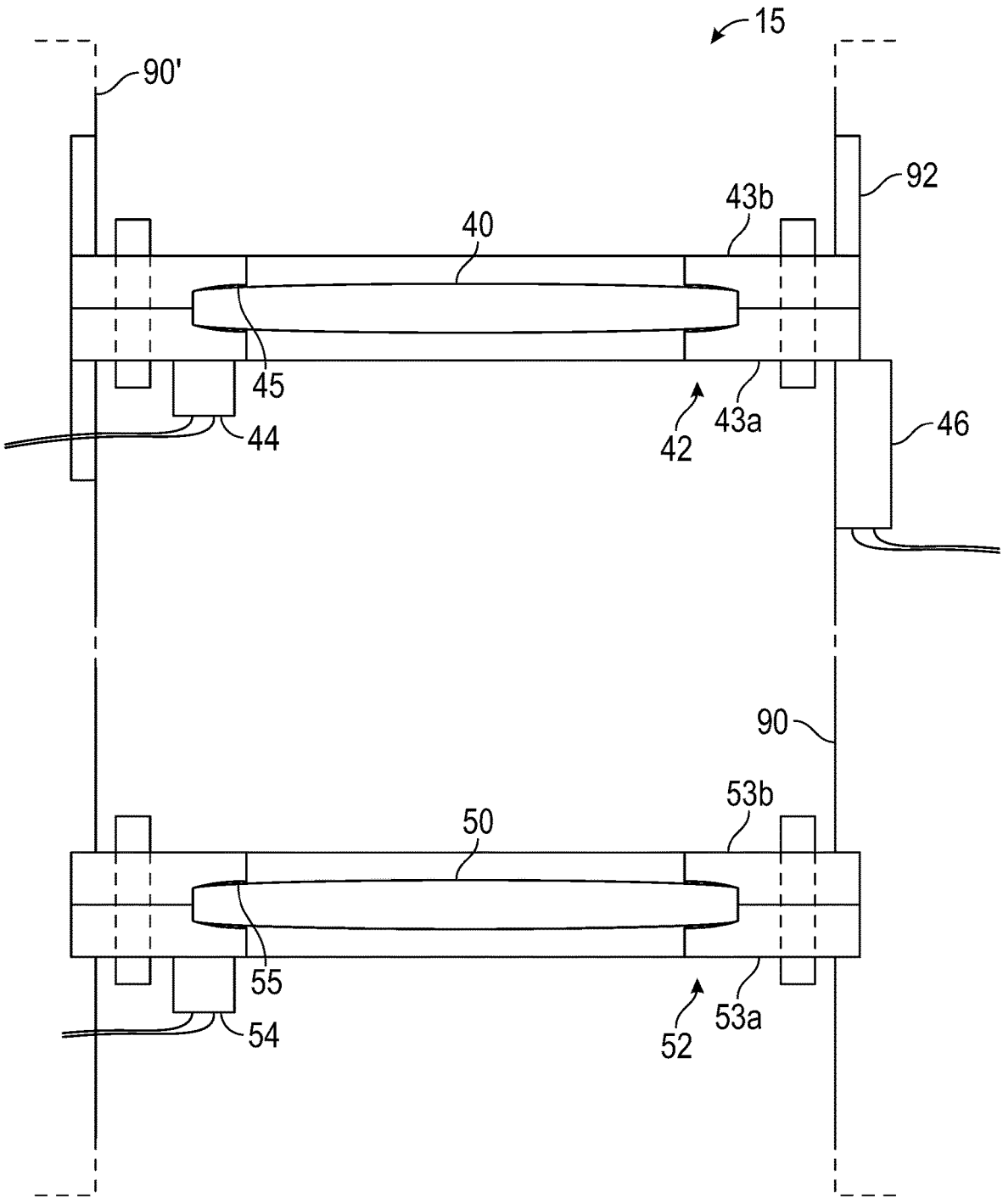


FIG. 2C

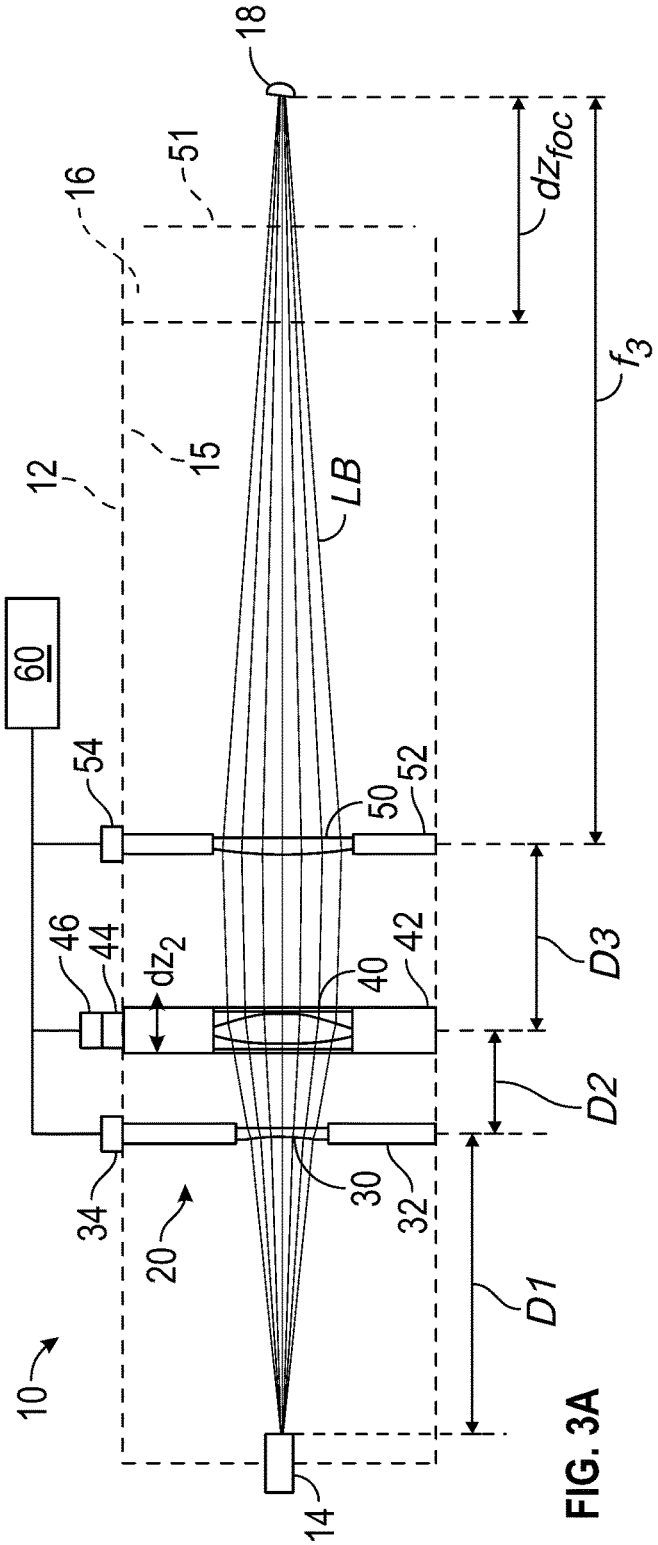

18
51
16
dz_{foc}
f_3
12
15
LB
60
54
50
52
46
44
dz_2
40
42
D3
D2
34
30
32
20
D1
10
14

FIG. 3A

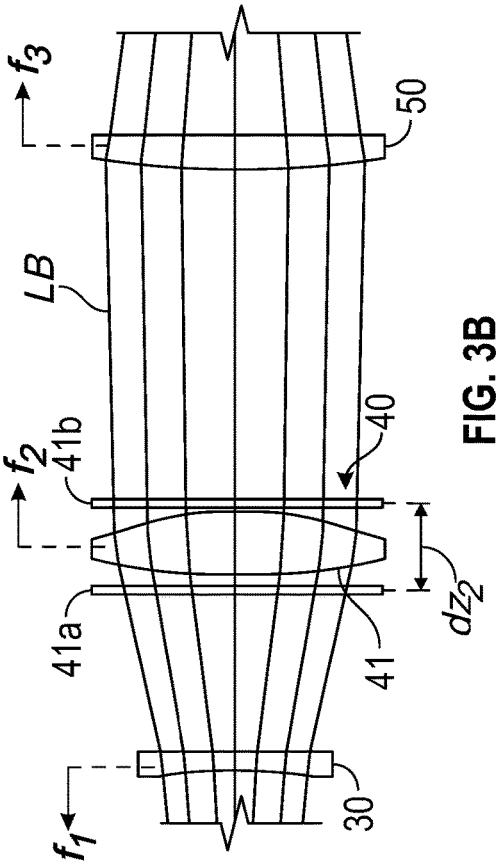

f_3
50
LB
f_2
41b
40
dz_2
41a
41
f_1
30

FIG. 3B

DYNAMIC FOCUS FOR LASER PROCESSING HEAD

FIELD OF THE DISCLOSURE

[0001] The present disclosure is directed to a high-power laser processing head configured to adjust its focal position dynamically while resisting contamination of the lenses and providing a large working distance.

BACKGROUND OF THE DISCLOSURE

[0002] A laser processing head, such as used for cutting or material processing, can provide a high-power laser beam of 10 kw and even up to 30 kW. Due to the working conditions, the high-power head is preferably resistant to contamination, which can damage the optics in the head and can undermine the laser's performance. In some laser processing heads, the focus position of the laser beam from the head can be adjusted along the optical axis.

[0003] Although existing arrangements to adjust the focus position may be effective, the configurations may not be well suited for harsh working conditions, high-power laser energy, exposure to contamination, and other issues that can damage the optics in the head and undermine the laser's performance. In addition, changing the focus position is a time critical operation and should be performed as quickly as possible. This can complicate the implementation of some arrangements.

[0004] The subject matter of the present disclosure is directed to overcoming, or at least reducing the effects of, one or more of the problems set forth above.

SUMMARY OF THE DISCLOSURE

[0005] A laser processing head is used to transmit a laser beam that is input along an optical axis of the head, to a process zone, the head comprises first, second and third lens elements, which can each include one or more lenses or optical elements. The first lens element is disposed in the head and images the laser beam from the input with a first focal length along the optical axis. The first focal length can be negative or can be positive for magnifications of about 6× or greater. The second lens element is disposed in the head and has an actuator arranged therewith. The second lens element images the laser beam from the first lens with a second (positive) focal length along the optical axis. The actuator is configured to move the second lens in the head along the optical axis over a variable lens distance relative to the first lens. The third lens element is disposed in the head and images the laser beam from the second lens element with a third (positive) focal length along the optical axis to a focal point. The third focal length is greater than the second focal length, and the focal point is variable along the optical axis over a variable focal distance in relation to the variable lens distance.

[0006] The head can comprise a housing having an input and an output for the laser beam along the optical axis. The first and third lens elements are fixedly mounted in the housing, whereas the second lens element is movably mounted in the housing in connection with the actuator.

[0007] The foregoing summary is not intended to summarize each potential embodiment or every aspect of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1**A illustrates a laser delivery system according to the present disclosure.

[0009] FIGS. **1**B-**1**C schematically illustrate a laser processing head according to the present disclosure.

[0010] FIG. **2**A illustrates a schematic view of a laser processing head having a lens arrangement according to the present disclosure.

[0011] FIG. **2**B illustrates a detail of the lens arrangement in FIG. **2**A.

[0012] FIG. **2**C schematically illustrates a lens, a fixed mounting structure, and a cooling element of the present disclosure and schematically illustrates a lens, a movable mounting structure, a cooling element, and an actuator of the present disclosure.

[0013] FIG. **3**A illustrates schematic view of a laser processing head having another lens arrangement according to the present disclosure.

[0014] FIG. **3**B illustrates a detail of the lens arrangement in FIG. **3**A.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0015] FIG. **1**A illustrates a laser delivery system **80** according to the present disclosure. A laser source **82** generates high-power laser light that is propagated along a fiber optic cable **84** to a laser processing head **10**. The laser source **82** can be an suitable multi-mode or a single-mode laser depending on the laser power required.

[0016] The laser processing head **10** can be moved relative to a workpiece WP and/or can have the workpiece WP moved relative to it. For example, the laser processing head **10** can be moved by a gantry system, robotic arm, or other apparatus **86** used in the art. Internally, the laser processing head **10** includes optics to focus the laser energy in a laser beam LB to the workpiece WP to achieve cutting, brazing, welding, additive manufacturing, or some other lasing process.

[0017] FIG. **1**B schematically illustrates the laser processing head **10** according to the present disclosure. The laser processing head **10** includes a housing **12** for internal optics of a lens arrangement **20**. A receiver **85** at one end of the housing **12** can couple to the laser cable **84**, which conducts the laser light into the head **12**. An output **16** at the other end of the housing **12** allows a focused laser beam LB to be emitted from the housing **12** for achieving the purposes of the lasing operation, such as welding, additive manufacture, cutting, etc. To protect the internal optics inside the housing **12**, the output **16** can include a cover slide that acts as a transparent pressure window.

[0018] During operation, the input **14** (e.g., a high-power laser delivery fiber) in the head **10** emits a high-power laser beam LB in the housing **12**. In a general sense, a fixed input lens element **30** images the laser beam LB to a second, movable lens element **40**. In turn, the movable lens element **40** images the laser beam LB to a fixed focusing lens element **50**, which focuses the laser beam LB to pass through the protective cover slide of the output **16**. The beam LB is focused to a focal point or spot **18**, and the laser beam LB then impinges on or near the workpiece WP or some other desired process zone. (Depending on the laser process and the type of head **10**, the focal point **18** can be located at or near the workpiece WP, inside a nozzle **17** as in FIG. **1**C

attached to the head, or at some other location along the optical axis that suites the process at hand.)

[0019] In a more technical sense, the first lens element **30** creates an image (typically a virtual image), which is de-magnified and serves as an object for the second lens element **40**. In turn, the second lens element **40** images this de-magnified object to close to infinity. Therefore, when the second lens element **40** is in a neutral position, the image created by the second lens element **40** is at infinity. Finally, the last lens element **50** images this new object from infinity to the focal plane of the last lens element **50**, forming the focal point **18** for the materials processing operation.

[0020] Movement of the second element **40** moves the virtual object for the last lens element **50** away from infinity and therefore results in a movement of the final focal point **18**. Demagnification of the intermediate object after first lens element **30** means, that for a given total magnification of the head **10**, the remaining optics consisting of second and third lens elements **40**, **50** must have a larger magnification M2. Moving the second lens element **40** is mostly equivalent to moving the virtual object. First order optics shows that this movement is translated to a movement of the focal point **18** by roughly M2*M2, meaning that the movement of the second lens element **40** can be much smaller than the resulting movement of the spot **18**.

[0021] The laser processing head **10** can be used in a laser process in which the focal point **18** of the laser beam LB can be adjusted in the optical axis (i.e., Z-axis) relative to a workpiece WP for laser cutting, welding, soldering, surface treatment, tactile brazing, additive manufacturing processes, and the like. The movable lens element **40** permits the focal point **18** of the laser beam LB to be adjusted along the optical axis. In this way, the focal point **18** can be adjusted to the different contours of the workpiece WP or to different process zones needed for the lasing process. This adjustment along the optical axis Z can be performed while the head **10** remains stationary relative to the workpiece WP, while the head **10** moves in the X-Y plane relative the workpiece WP, while the head **10** is also moved along the Z-axis, or when a combination of such movements are performed.

[0022] FIG. **2**A illustrates more detail of a laser processing head **10** having a lens arrangement **20** according to the present disclosure. As noted, the head **10** focuses a laser beam LB to the focal point **18**. The focal point **18** can be adjusted relative to a focal plane **51**, which can be outside or inside the head **10** (or component thereof, such as a nozzle) depending on the lasing process. As only schematically shown here, the head **10** includes the housing **12** having the input **14** and the output **16** for the laser beam LB to pass along the optical axis A. In an interior **15**, the housing **10** includes the lens arrangement **20** with lens elements **30**, **40**, **50** disposed in the interior **15** of the housing **10**. In general, each of the lens elements **30**, **40**, **50** can include one lens, but they may also comprise a group of multiple lenses, composite lenses, or optical elements in the form of a lens group. (Reference to a lens may be used herein for the purpose of description, but it should be appreciated that each of the lens elements **30**, **40**, **50** can include one or more lenses or optical elements.) In one arrangement, the first, input lens element **30** is mounted in the housing's interior **15** using a fixed mounting structure **32**; the second, intermediate lens element **40** is mounted in the housing's interior **15** using a movable mounting structure **42**; and the third, focusing lens element **50** is mounted in the housing's interior **15** using a fixed mounting structure **52**.

[0023] The input mounting structure **32** is in a fixed position in the housing **10** so that the input lens element **30** is set at a first distance D_1 from the input **14**, which can include an optical fiber input suited to the high-power laser of the head **10**. The intermediate mounting structure **42**, by contrast, is mounted in conjunction with an actuator **46** operable to adjust the position of the intermediate lens element **40** in the housing **10** along the optical axis A. In this way, the second distance D_2 of the intermediate lens element **40** relative to the input lens element **30** can be controlled and varied. Finally, the output mounting structure **52** is in a fixed position in the housing **10** so that the output lens element **50** is set at a distance (D_2+D_3) from the input lens element **30**. The output lens element **50** focuses the laser beam LB to a focal point **18** that can be varied a variable distance dz_{foc} relative to a focal plane **51** of the output lens element **50**. (In FIG. **2**A, the focal point **18** is shown beyond the focal plane **51** of the output lens element **50** at the most distanced boundary (right in the drawing) of the range for the variable distance dz_{foc}. This means also that the intermediate lens element **40** is shown in FIG. **2**A at its farthest boundary (left in the drawing) of its movement range from the output lens element **50**. The laser beam LB between the intermediate and output lens elements **40** and **50** diverges. Although not shown, when the intermediate lens element **40** is positioned at its closest boundary (right in the drawing) of its movement range toward the output lens element **50**, then the focal point **18** would be inside the focal plane **51** of the output lens element **50** at the shortest boundary (left in the drawing) of the range for the variable distance dz_{foc}. The laser beam LB between the intermediate and output lens elements **40** and **50** would converge in this instance.)

[0024] In general, the focal point **18** can be beyond the housing's output **16**, but the focal point **18** does not necessarily have to be outside of the housing "proper", depending on the type of housing used and any additional components used on the head **10**. In two-dimensional flat sheet cutting, for example, the focus of the laser beam LB is controlled with respect to a cutting gas nozzle affixed on an end of the head **10**. During use, the focal point **18** can be both inside, at, or outside this nozzle. The actuator **46** on the intermediate lens element **40** can adjust/control the third distance D_3 of the intermediate lens element **40** to the output lens element **50** in direct offset to the adjustment to the second distance D_2.

[0025] As also shown in the detail of the lens arrangement **20** in FIG. **2**B, the input lens element **30** images the laser beam LB from the input **14** with a first, input focal length f_1 along the optical axis A. In one configuration, the input focal length f_1 is negative. The intermediate lens element **40** images the laser beam LB from the input lens element **30** with a second, intermediate focal length f_2 along the optical axis A. The intermediate focal length f_2 is positive. As noted, the actuator **46** is configured to move the intermediate lens element **40** in the head **10** along the optical axis A at a variable lens distance dz_2 relative to the input and output lens elements **30**, **50**.

[0026] In contrast to the previous configuration, the input focal length f_1 can instead be positive when the head **10** has a very large magnification, such as for magnifications greater than about 6×. The range of parameters disclosed

herein for the head **10** can include these large magnifications, which may be useful for cutting processes.

[0027] The output lens element **50** images the laser beam LB from the intermediate lens element **40** with a third, output focal length f_3 along the optical axis A to the focal point **18**. The output focal length f_3 is positive. The focal point **18** is variable over a variable focal distance dz_{foc} along the optical axis A in relation to the increase/decrease of the variable lens distance dz_2 of the intermediate lens element **40** relative to the input and output lens elements **30**, **50**. Moving the lens element **40** towards the output lens element **50** will move the focal point **18** toward the output lens element **50**, and moving the intermediate lens element **40** away from the output lens element **50** will move the focal point **18** away from the output lens element **50**.

[0028] Because the laser head **10** is used for high-power lasing, cooling of one or more of the lens elements **30**, **40**, and **50** can increase imaging performance of the head and can help prevent adverse effects. Preferably, the head **10** includes at least one cooling arrangement disposed in the housing **10**. The at least one cooling arrangement is disposed in thermal communication with at least one of the lenses **30**, **40**, **50** and is configured to cool the associated lens element **30**, **40**, **50**.

[0029] The outer stationary lens elements **30** and **50** allow for good sealing of an optics room **25** in the head **10** and can be cooled efficiently. The optics room **25** for the lens elements **30**, **40**, **50** is preferably sealed and is preferably as small as possible, which facilitates cooling and sealing. When using lens material having high thermal conductivity such as sapphire (Al_2O_3), zinc sulfide (ZnS), these lens elements **30** and **50** are very resistant to contamination. Due to the small movement range of the intermediate movable lens element **40** made possible for the reasons noted above, active cooling is at least easier for it as well. In addition, the small movement range allows for a very fast adjustment of the focal point **18**.

[0030] At least in one configuration, for example, the input lens element **30** includes a cooling element **34** because this input lens element **30** is subject to more concentrated imaging of the laser beam LB from the input **14** and may be subject to the most heating per unit surface area. In another configuration, the first and third lenses **30**, **50** include cooling elements **34**, **54** configured to cool the associated lens elements **30**, **50** because these lens elements **30**, **50** are mounted in a fixed position in the housing **12**, making the cooling elements **34**, **54** easier to incorporate into the mounting structure **32**, **52**.

[0031] In yet another configuration, each of the lens elements **30**, **40**, **50** includes a cooling element **34**, **44**, **54** disposed in thermal communication therewith and configured to actively cool the associated lens element **30**, **40**, **50**. In general, the cooling elements **34**, **44**, **54** can include thermoelectric cooling elements on the mounting structures **32**, **42**, **52**.

[0032] Because the input and output lens elements **30**, **50** are fixed in the head **10** with mounting structures **32**, **52**, any desired sensors can be more readily used for monitoring these stationary lens elements **30**, **50**. Such sensors (not shown) can include a temperature sensor, a viewing sensor, or other form of detector to monitor the health of the lens element **30**, **50**. Such a sensor requires certain wiring or cabling that can be simplified for these fixed mounts of the mounting structures **32**, **52**. Moreover, the input and output lens elements **30**, **50** being fixed in the head **10** with the mounting structures **32**, **52** simplifies how the associated cooling element **34**, **54** can be incorporated into (onto) the mounting structure **32**, **52** and adjacent housing **12**. A number of mounting structures **32**, **42**, **52** can be used for the lens elements **30**, **40**, **50**. For example, the fixed position mounting structures **32**, **52** can use components such as lens tubes, screw rings, and clamped mounts. The lens elements **30**, **40**, **50** can be mounted with seals and can be held in place in the mounting structures **32**, **42**, and **52**.

[0033] Briefly, FIG. **2**C schematically illustrates details of a fixed lens element (e.g., **50**), a fixed mounting structure **52**, and a cooling element **54** of the present disclosure. Here, the mounting structure **52** is a clamp-type mount with clamp elements **53***a-b* affixed together and holding the lens element **50** therein. Sealing at **55** can be used to fix the lens element **50** in the clamp elements **53***a-b*. The mounting structure **52** can be affixed or mounted in the interior **15** of the housing using conventional features (not shown). For example, a module element **90** may be mounted in the interior **15**.

[0034] The cooling element **54**, such as one or more thermoelectric cooling elements, can be disposed on (or disposed in) the clamp type mount **52**. As will be appreciated with the benefit of this example, a number of mounting structures and arrangements can be used to mount the fixed lenses (**30**, **50**) in the head's housing.

[0035] Briefly, FIG. **2**C also schematically illustrates details of the movable lens (e.g., intermediate lens element **40**), a movable mounting structure **42**, a cooling element **44**, and an actuator **46** of the present disclosure. Here, the mounting structure **42** is a clamp-type mount with clamp elements **43***a-b* affixed together and holding the lens element **40** therein. Sealing at **44** can be used to fix the lens element **40** in the clamp elements **43***a-b*. The mounting structure **42** can be mounted in or on a rail, track, or other guide in the interior **15** of the housing using conventional features. For example, a module element **90'** may be mounted in the interior **15** and can include the guide **92**. This module element **90'** can be part of the same module used for output lens and/or the input lens.

[0036] The cooling element **44**, such as one or more thermoelectric cooling elements or water cooling arrangements, can be disposed on (or disposed in) the clamp type mount **42**. The actuator **46**, which can include one or more elements, can be mounted with respect to the housing's interior **15** in conjunction with the mount **42** on one or more locations or sides of the mount **42**. The actuator **46** can move the mount **42** along the guide **92**, track or the like to adjust the position of the lens element **40**. As will be appreciated with the benefit of this example, a number of mounting structures and arrangements can be used to mount the movable lenses (**40**) in the head's housing.

[0037] Because the intermediate lens element **40** as shown in FIGS. **2**A-**2**B is movable at a small variable lens distance dz_2 in the head **10** by operation of the actuator **46**, any desired sensors (not shown) can still be used for monitoring this lens element **40** even with its movement. Such sensors can include a temperature sensor, a viewing sensor (i.e., photodiodes), or other form of detector to monitor the health of the lens. Such a sensor requires certain wiring or cabling that can be simplified for the mounting structure **42** that moves only the small variable lens distance dz_2. Moreover, the intermediate cooling element **44** can still be incorporated

into the mounting structure **42**, and the cooling element **44** and mounting structure **42** can be moved by the actuator **46**.

[0038] As shown in FIG. **2**A, the actuator **46** for the intermediate lens element **40** is controlled by a controller **60**. Preferably, the actuator **46** allows for highly dynamic control to achieve the small movement dz_2 of the intermediate lens element **40** along the optical axis A. The actuator **46** can include one or more unidirectional mover, bi-directional mover, motor, solenoid, piezoelectric element, linear motor, voice coil motor, or other device having precise, controlled movements over a short distance. Movement over the small distance dz_2 can be achieved with one actuator **46** capable of bi-directional movement. Alternatively, opposing unidirectional actuators **46** can be disposed on opposing sides of the mounting structure **42** to achieve the bi-directional movement. The actuator **46** can be combined with bearings, joints, guides, biasing elements, springs, and any other necessary mechanical features to control movement. Overall, the actuator **46** can use any form of mechanical or electromechanical actuation that has fine precision and accuracy and that can accommodate cooling. Any of the cooling elements **34**, **44**, **54** in the head **10** can also be controlled by the controller **60**, which can be a microcontroller specifically associated with the head **10** or can be part of a larger control system for a laser delivery system (e.g., **80**: FIG. **1**A).

[0039] The controller **60** operates the actuator **46** to adjust the position of the intermediate lens element **40** along the small variable lens distance dz_2, which in turn adjust the focus position **18** of the laser beam LB along the larger variable focal distance dz_{foc}. As noted, this focus position **18** can be adjusted relative to surfaces of a workpiece, while the head **10** is stationary relative to the workpiece, while it moves relative to the workpiece, etc.

[0040] Adjusting the focus position **18** can be used in a number of ways during laser processing of the workpiece. While the head **10** is stationary relative to the workpiece, movement of the intermediate lens element **40** along the small variable lens distance dz_2 can adjust the focus position **18** with respect to surfaces of the workpiece. While the head **10** is moved along an axis consistently with respect to the workpiece or vice versa, movement of the intermediate lens element **40** along the small variable lens distance dz_2 can adjust the focus position **18** with respect to surfaces of the workpiece. (In other words, the focal point **18** can be moved relative to the head **10** during operations. If the head **10** is stationary, the focal point **18** can be moved relative to both workpiece and head **10**. If the head **10** is moved relative to the workpiece and if at the same time focal point **18** is moved relative to the head **10**, the focal point **18** can be kept stationary with respect to the workpiece while moving the head **10**. These and other combinations of movement can be used.) Moreover, using the actuator **46** suitable for the small range of movement over the variable lens distance dz_2, the intermediate lens element **40** can be further operated to have high frequency oscillations in the head **10**, which produces high frequency oscillations of the focal point **18** along the optical axis A. These and any other combinations of adjustments can be used by actuating the intermediate lens element **40**.

[0041] For optical control and as best shown in FIG. **2**B, the input focal length f_1 is negative; the intermediate focal length f_2 is positive, and the output focal length f_3 is positive. The output optical length f_3 is greater than the intermediate focal length f_2. Typically, a movement range of the intermediate lens element **40** can be less than about 15 mm, and the spot movement can be larger than 60 mm. In general, the third focal length f_3 can greater than the second focal length f_2 such that a ratio of $|f_3/f_2|$ can be at least two (2).

[0042] With this optical arrangement, the laser beam LB after the first two lenses **30**, **40** is close to being collimated. The input lens element **30** creates a smaller virtual image of the fiber tip of the input **14**, and this smaller virtual image is then imaged by the intermediate lens element **40** and the output lens element **50** towards the workpiece.

[0043] Due to the reduced virtual image from the input lens element **30**, this initial part of the laser beam LB has a large magnification. Therefore, the ratio of the output focal length f_3 relative to the intermediate focal length f_2 is so large that a very small movement of the intermediate lens element **40** along the variable lens distance dz_2 results in a large movement of the focus position **18** along the variable focal distance dz_{foc}. For typical adjustment required to focus position that would be used in 2-D sheet metal cutting, the required lens movement distance dz_2 can be on the order of a few millimeters, allowing for highly dynamic adjustments of the laser beam's focal point **18**.

[0044] In general, the input lens element **30** can be composed of sapphire (Al_2O_3), zinc sulfide (ZnS), or other material having "high" thermal conductivity (e.g., glass having thermal conductivity at least as great as sapphire, or zinc sulfide). In general, the thermal conductivity of aluminum oxide or sapphire varies based on temperature and orientation, but can be about 24 W/(m*K). In general, the thermal conductivity of zinc sulfide varies based on temperature, but can be about 16.7 W/(m*K) at 20° C. These values are only meant to represent possible examples for the high thermal conductivity desired for the lens material. The output lens element **50** can also be composed of sapphire, zinc sulfide, or other material (e.g., glass having high thermal conductivity of at least as great as sapphire, or zinc sulfide).

[0045] For its part, the intermediate lens element **40** can be composed of sapphire, zinc sulfide, or other material (e.g., glass having high thermal conductivity of at least as great as sapphire, or zinc sulfide). Instead of being composed of sapphire, or zinc sulfide, however, normal fused silica may be used in some or all of the lenses **30**, **40**, **50** to reduce costs or improve manufacturability with the tradeoff of reduced contamination resistance.

[0046] At least one of the lens elements **30**, **40**, **50** can be aspherical. In one configuration, the input lens element **30** is a spherical lens; the intermediate lens element **40** is an aspherical lens; and the output lens element **50** is either spherical or aspherical. For example, either the intermediate lens element **40**, the output lens element **50**, or both can be aspheric to reduce imaging error. If sapphire or zinc sulfide is used for both of the input and output lens elements **30**, **50**, the lens arrangement **20** having a single aspheric lens can be sufficient.

[0047] The focal lengths, the distances, and the optical diameters for the lens elements **30**, **40**, **50** depends on the size of the housing **12**, the optical power of the laser beam LB, and other operational characteristics. The configuration can be generally characterized by a number of interrelated relationships discussed below.

[0048] As noted, the input lens element **30** has a first focal length f_1; the intermediate lens element **40** has a second focal length of f_2; and the output lens element **50** has a third

focal length f_3. The lens' variable distance dz_2 represents the movement range for the intermediate lens element **40**, and the expanse dz_{foc} is the movement range of the focal point **18** around the focal plane of the output lens element **50**. Based on these variables, the focal length f_2 of the intermediate lens element **40** can be characterized by:

$$f_2 = \sqrt{\frac{dz_2}{dz_{foc}}}$$

[0049] The focal length f_3 of the output lens element **50** corresponds to about the working distance of the head **10**. The collimated focal length f_{col} grouped for the input lens element **30** relative to the intermediate lens element **40** at the neutral position is characterized by:

$$f_{col} = \frac{f_3}{M}$$

[0050] The variable M is the magnification involved. Given the distance D_2 between input lens element **30** and the intermediate lens element **40**. The focal length f_1 of the input lens element **30** is characterized by:

$$f_1 = \frac{1 - \frac{D_2}{f_2}}{\frac{1}{f_{col}} - \frac{1}{f_2}}$$

[0051] Finally, the distance D_1 from the input **14** of the fiber tip to the input lens element **30** is characterized by:

$$D_1 = \frac{1 - \frac{D_2}{F_2}}{\frac{1}{f_1} + \frac{1}{f_2} - \frac{D_2}{f_1 * f_2}}$$

[0052] Actual values for these parameters of the head **10** will depend on the particular implementation. Below is a brief discussion of some typical parameter values.

[0053] In general, the third focal distance f_3 (i.e., working distance) can be greater than or equal to 200-mm and can be about 200-mm to 500-mm, resulting in a large working distance and large beam diameters for large laser power operation of the head **10**. The variable focal distance dz_{foc} that the focal point **18** can vary along the optical axis can be from about 40-mm to 90-mm, depending on the requirements for cutting or other laser processes. The variable lens distance dz_2 of the intermediate lens element **40** can be in the range of about 5-mm to 15-mm. This gives the adjustments a high dynamic range and allows the intermediate lens element **40** to be more easily cooled. Overall, the variable focal distance dz_{foc} can be about 2 to 18 times greater than the variable lens distance dz_2.

[0054] Continuing with the examples, the distance D_2 between the input lens element **30** and the intermediate lens element **40** can be about 20-mm to 100-mm so that the optics can be more compact in the head **10**. The first distance D_1 from the fiber tip **14** to the input lens element **30** can be greater than 50-mm so that the input lens **20** is not located too close to the input fiber **14**. These values discussed above are only presented as an example and can vary from one implementation to another depending on the characteristics.

[0055] In one particular configuration, the lens arrangement **20** can be sized and used for the head **10** having the housing **12** with an overall distance of the laser beam LB of about 480-mm from the input **14** to the output (e.g., nozzle) of the head **10**. The input distance D_1 can be about 100-mm; the distance between the input lens element **30** and the outlet lens element **50** (i.e., distances D_2+D_3) can be about 80-mm; the distance between input lens element **30** and movable lens element **40** (distance D_2) can be about 30 mm; and the working distance f_3 can be 300-mm. Thus, when the working distance f_3 is about 300-mm, the magnification M at a Z position −20-mm can be about 1.73×, whereas the magnification M at a Z position +40-mm can be about 2.14×.

[0056] With these dimension, the beam diameter of the laser beam LB at the input lens element **30** can be about 30-mm, and the beam diameter at the intermediate lens element **40** can be about 40-mm with typical cutting laser sources with a beam parameter product (BPP) of about 4 mm*mrad. As noted, the laser beam LB is practically collimated between the intermediate lens element **40** and output focusing lens element **50**.

[0057] With these dimension, the intermediate lens element **40** can be varied along a variable lens distance dz_2 of about ±2 mm. As already noted, the small movement needed for the intermediate lens element **40** allows the optics room **25** to be more readily sealed from the mechanics of the actuator **46**, to be more readily cooled, and to be easily fine-tuned.

[0058] FIG. **3**A illustrates a laser processing head **10** having another lens arrangement **20** according to the present disclosure, and FIG. **3**B shows a detail of the lens arrangement **20** in FIG. **3**A. This head **10** and lens arrangement **20** is similar to the previous configuration of FIGS. **2**A-**2**B so that similar reference numbers are used for comparable components. Description of these comparable components is reincorporated here, but may not be repeated below.

[0059] As before, the lens arrangement **20** has three lens elements **30**, **40** and **50**, which can each include one lens or group of lenses or optical elements. The first, input lens element **30** has a negative focal length f_1, is actively cooled, is stationary in the optical axis A, and is composed of a material having a high thermal conductivity. For example, the input lens element **30** can be composed of sapphire, or zinc sulfide. (As noted previously, the first focal length can be positive when the head **10** has a magnification greater than about 6× in some configurations.) The third, output lens element **50** has a positive focal length f_3, is actively cooled, is stationary in the optical axis A, and is composed of a material with a high thermal conductivity as noted herein.

[0060] As before, the second, intermediate lens element **40** has a positive focal length f_2 and can be moved along the optical axis A using the actuator **46**. The intermediate lens element **40** can also be cooled with a cooling element **44**. Again, the intermediate mounting structure **42** is mounted in conjunction with the actuator **44** operable to adjust the position of the intermediate lens element **40** in the housing **10** along the variable lens distance dz_2 of the optical axis A.

[0061] The intermediate lens element **40**, however, includes a number of differences in contrast to the previous arrangements. In particular, the intermediate lens element **40**

is mounted in the mounting structure **42** as a lens group further including a pair of cover slides **41***a-b*. The lens **41** for the element **40** in this configuration can be composed of a different material than the other configurations and can, for example, be composed of a fused silica. This intermediate lens **41** of the element **40** is sandwiched between the cover slides **41***a-b*, which can be composed of sapphire, zinc sulfide, or other material of "high" thermal conductivity noted herein. This configuration may make tradeoffs as to size, weight, and performance compared to the previous configuration, the lens arrangement **20** here can still be kept cleaner from contamination. The intermediate lens **41** can be composed of fused silica, which is a less expensive material, and can be aspherical. The aspherical, fused silica lens **41** sandwiched between the two sapphire cover slides **41***a-b* therefore provides for reduced manufacturing costs, yet can keep the configuration clean from contamination. The cover slides **41***a-b* seal the aspherical lens **41** in a small volume to keep the lens **41** clean. Contamination from outside is handled by the contamination resistant sapphire, or zinc sulfide of the cover slides **41***a-b*.

[0062] The foregoing description of preferred and other embodiments is not intended to limit or restrict the scope or applicability of the inventive concepts conceived of by the Applicants. It will be appreciated with the benefit of the present disclosure that features described above in accordance with any embodiment or aspect of the disclosed subject matter can be utilized, either alone or in combination, with any other described feature, in any other embodiment or aspect of the disclosed subject matter.

What is claimed is:

1. A laser processing head to transmit a laser beam that is input along an optical axis of the head, to a process zone, the head comprising:

a first lens element disposed in the head and imaging the laser beam from the input with a first focal length along the optical axis;

a second lens element disposed in the head and having an actuator arranged therewith, the second lens element imaging the laser beam from the first lens with a second focal length along the optical axis, the second focal length being positive, the actuator being configured to move the second lens in the head along the optical axis over a variable lens distance relative to the first lens; and

a third lens element disposed in the head and imaging the laser beam from the second lens element with a third focal length along the optical axis to a focal point, the third focal length being positive and being greater than the second focal length, the focal point being variable along the optical axis over a variable focal distance in relation to the variable lens distance.

2. The head of claim **1**, wherein the first lens element is disposed at a fixed position in the head.

3. The head of claim **1**, wherein the first lens element is a spherical lens.

4. The head of claim **1**, wherein: the first focal length is negative, the first lens element images the input as a de-magnified object for the second lens element, the second lens element images the de-magnified object toward infinity as a resulting object, and the third lens element images the resulting object to a focal plane of the third lens element.

5. The head of claim **1**, wherein the first focal length is positive, and wherein the head has a magnification of about 6× or greater.

6. The head of claim **1**, wherein the first lens element comprises a lens composed of (i) fused silica, (ii) sapphire, (iii) zinc sulfide, or (iv) material having a thermal conductivity of at least as great as sapphire or zinc sulfide.

7. The head of claim **1**, wherein the second lens element comprises a lens composed of (i) fused silica, (ii) sapphire, (iii) zinc sulfide, or (iv) material having a thermal conductivity of at least as great as sapphire or zinc sulfide.

8. The head of claim **1**, wherein the second lens element comprises a fused silica lens disposed between a front cover slide and a back cover slide; wherein the front and back cover slides are composed of (i) sapphire, (ii) zinc sulfide, or (iii) material having a thermal conductivity of at least as great as sapphire or zinc sulfide.

9. The head of claim **1**, wherein the third lens element is disposed at a fixed position in the head.

10. The head of claim **1**, wherein the third lens element comprises a lens composed of (i) fused silica, (ii) sapphire, (iii) zinc sulfide, or (iv) material having a thermal conductivity of at least as great as sapphire or zinc sulfide.

11. The head of claim **1**, wherein the first lens element and third lens element are each disposed at a fixed position in the head, and wherein the first and third lens elements define an optics room in the head, the optics room being sealed and enclosing the second lens element.

12. The head of claim **1**, wherein the second lens element, the third lens element, or both comprise an aspherical lens configured to reduce imaging error.

13. The head of claim **1**, further comprising a cooling arrangement having at least one cooling element disposed in thermal communication with at least one of the first, second, and third lens elements and being configured to actively cool the at least one lens element.

14. The head of claim **1**, wherein the actuator comprises one or more unidirectional mover, bi-directional mover, motor, linear motor, solenoid, voice coil, or piezoelectric element.

15. The head of claim **1**, wherein the third focal length is greater than the second focal length by a factor of at least two.

16. The head of claim **1**, wherein the variable focal distance is about 2 to 18 times greater than the variable lens distance.

17. A laser processing head to transmit a laser beam to a process zone, the head comprising:

a housing having an input and an output for the laser beam along the optical axis;

a first lens element fixedly mounted in the housing and imaging the laser beam from the input with a first focal length along the optical axis;

a second lens element movably mounted in the housing in connection with an actuator, the second lens element imaging the laser beam from the first lens element with a second focal length along the optical axis, the second focal length being positive, the actuator being configured to move the second lens element in the head along the optical axis at a variable lens distance relative to the first lens element; and

a third lens mounted in the housing and imaging the laser beam from the second lens element with a third focal length along the optical axis to a focal point, the third

focal length being positive and being greater than the second focal length, the focal point being variable along the optical axis at a variable focal distance in relation to the variable lens distance.

18. The head of claim **17**, further comprising at least one cooling arrangement disposed in the housing in thermal communication with at least one of the first, second, and third lens elements and configured to cool the at least lens element.

19. The head of claim **17**, wherein the first lens element is a spherical lens; and

wherein the second and/or third lens element is an aspherical lens configured to reduce imaging error.

20. The head of claim **17**, wherein:

the first focal length is negative; or

the first focal length is positive and the head has a magnification of about 6× or greater.

21. The head of claim **17**, wherein the second lens element comprises:

a lens composed of (i) sapphire, (ii) zinc sulfide, or (iii) material having a thermal conductivity of at least as great as sapphire or zinc sulfide; or

a fused silica lens disposed between a front cover slide and a back cover slide, the front and back cover slides being composed (i) sapphire, (ii) zinc sulfide, or (iii) material having a thermal conductivity of at least as great as sapphire, or zinc sulfide.

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